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Capstone Assignment

Case Study - 1 Processor Selection for a Smart Traffic Control System

1. Role of basic functional Units.

• ALU - Perform real time calculations like time intervals between vehicle detections, traffic flow rates countdown timers for signal changes.

• Control Unit - Manages the instruction sequence, interprets sensor interupts and coordinates state transitions.

• Memory Unit - Stores sensor data transfers, traffic pattern history and timing parameters, fast access is critical for real - time response.

• I/O Interface - Read from inductive loop sensors, cameras radar units etc. Output to traffic light actuators and communication network.

• Clock - Provides precise timing for signal phase durations and synchronisation between interactions.

2. RISC vs CISC:

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Risc is more suitable

- Real-time predictably - Fixed-length, single-cycle execution and deterministic timing with RISC

Power efficiency - RISC gives simpler decode logic and smaller pipeline stages.

Interrupt response - Quick Context Switch thus minimal save and restore.

CISC's variable instruction lengths introduce unacceptable worst case delays.

3. Instruction Throughput (IPC) and Performance Impact

For this system, consistent throughput matters more than peak IPC.

- IPC measures how many instructions complete per clock cycle. A steady IPC of 0.9 - 1.0 is better than bursty 1.5 IPC with stalls.

- Real time constraints - Sensor data must be processed every 10 - 50 ms. If IPC drops due to cache misses or hazards, the controller misses timing windows - potential intersection deadlock.

- Performance bottleneck is I/O and memory access not raw compute. A modest 50 - 100 MHz RISC core with high IPC is sufficient.

4. Processor Datapath Selection - Single-Cycle, Multi-cycle or Pipeline

For a smart traffic control system requiring real time sensor processing, quick response and moderate power, the choice of processor is critical

4.1. Single-cycle processor

In a single-cycle design, every instruction completes in one long cycle. The clock period is set by the slowest instruction.

Not suitable. The system needs to process multiple sensors within 10-50 ms - a single cycle processor would waste power and cannot meet real time demands.

4.2 Multi-Cycle processor

Here, instructions takes variable number of shorter cycles. A single ALU and memory are reused

Acceptable only for simple fixed timing interactions. Not ideal for smart systems with adaptive signals or V2X communication.

4.3. Pipelined processor - A pipeline processor overlaps multiple instructions. A 3-stage pipeline (Fetch → Decode → Execute) allows one instruction per cycle after filling.

A 3-Stage pipelined RISC processor at 80-120 MHz is recommended for this system

5. Pipeline Hazards and Mitigation in Traffic System

When using pipelined processor, 3 hazards can disrupt execution -

- Structural Hazards
  - Two instructions needs same hardware resource simultaneously. for example - Instruction fetch and data load sharing a single memory bus.
  - Simultaneously read from a sensor port and write an actuator port on same I/O bus.

mitigations - Use Harvard architecture - standard in embedded RISC cores.
Implement dual-ported I/O banks with independent request queues.

- Data Hazards
  - An instruction tries to read a register before a previous instruction writes it.

mitigations - Route ALU directly from execute to Decode stage thus eliminate most raw hazards without stalls.
- Insert one stall cycle directly for load-use hazards.
- Reorder independent instructions.

- Control Hazards

Pipeline fetches the next instruction before knowing whether a branch is taken. If taken, fetched

instructions must be flushed.

Mitigations

- Short pipeline - Branch penalty is only 1 cycle which is much better than 5+ stage pipeline.

- Static branch prediction - For traffic control, most branches stay in normal timing loop.

- Interrupt based emergency holding - Instead of pooling, use hardware based interrupts for emergencies.